

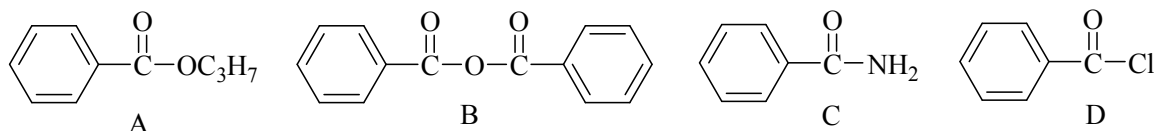


厦门大学《有机化学 B》课程

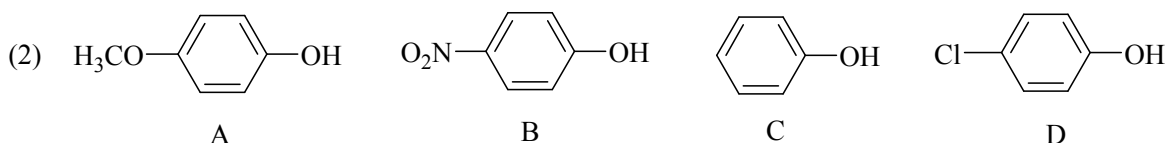
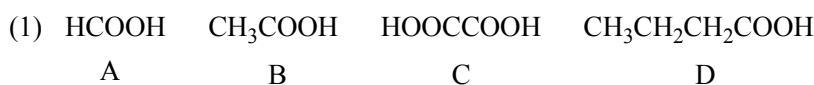
练习题(I)

一、回答下列问题：(10分)

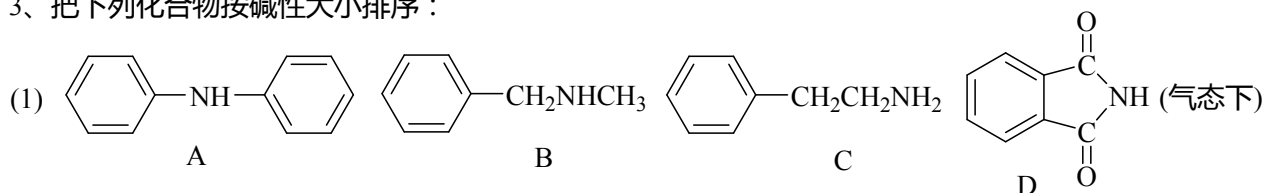
1、比较下列化合物发生水解反应的活性大小：



2、把下列化合物按酸性大小排序：



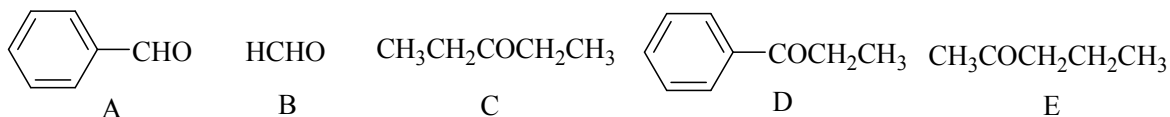
3、把下列化合物按碱性大小排序：



4、能与三氯化铁发生颜色反应的是下列哪类结构的化合物？

(A) 羟基结构 (B) 链醚结构 (C) 烯醇结构 (D) 芳醚结构

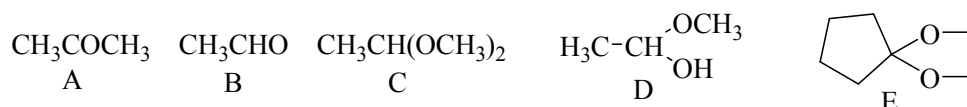
5、排出下列化合物进行亲核加成反应的难易次序：



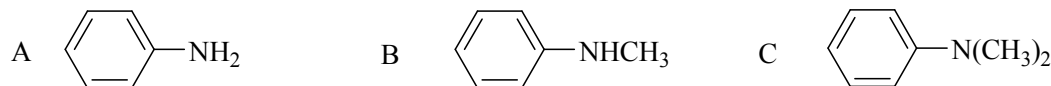
6、比较下列化合物的酯化反应速率：

(A) CH_3COOH (B) $(\text{CH}_3)_3\text{CCOOH}$ (C) $(\text{CH}_3)_2\text{CHCOOH}$ (D) $\text{CH}_3\text{CH}_2\text{COOH}$

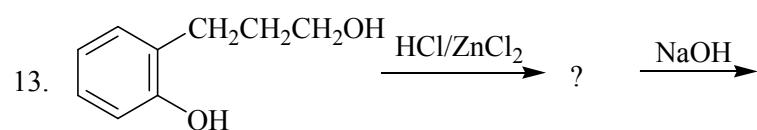
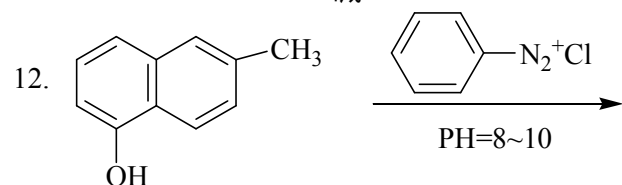
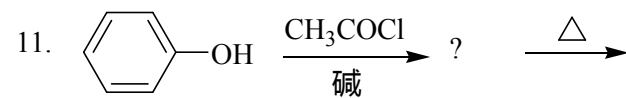
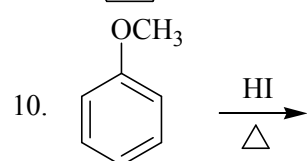
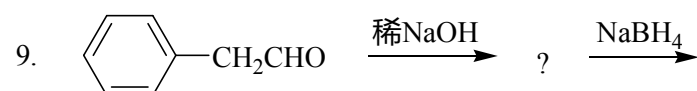
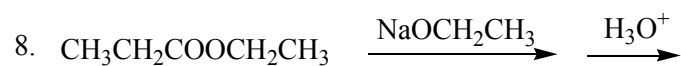
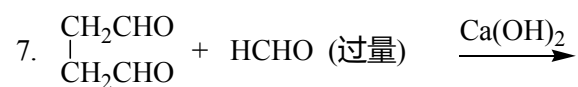
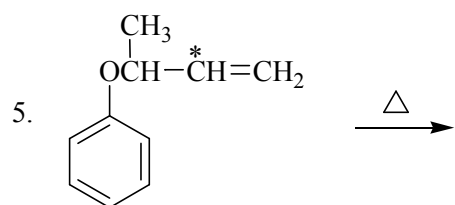
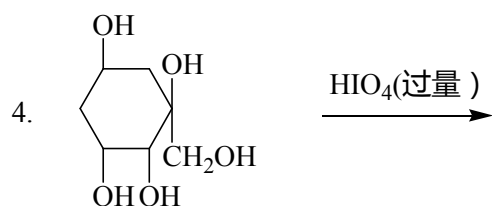
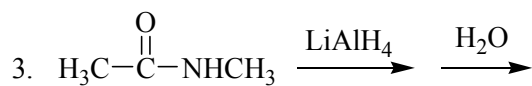
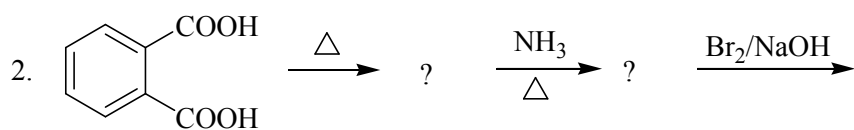
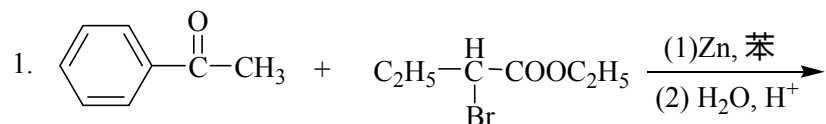
7、下列哪些化合物能发生银镜反应？

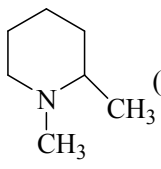
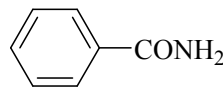
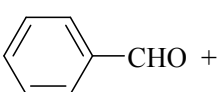
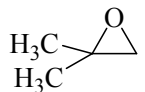


8、下列化合物分别在碱存在下与对甲苯磺酰氯反应，析出固体的是：



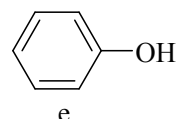
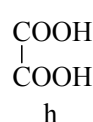
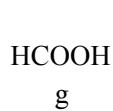
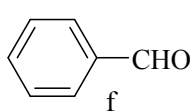
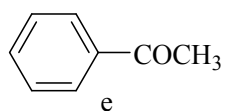
2、写出下列反应的主要反应产物（30分）

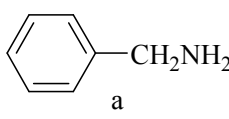
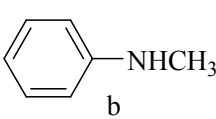
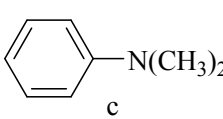
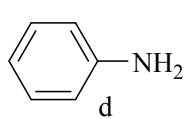
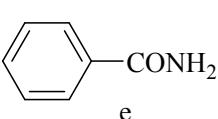


14.  $\xrightarrow{\text{CH}_3\text{NH}_2}$
15. $\text{CH}_3\text{COCH}_2\text{COOC}_2\text{H}_5 \xrightarrow[(2)\text{H}_3\text{O}^+]{(1)\text{浓碱}}$
16. $\text{HO}-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{COOH} \xrightarrow{\Delta}$
17.  $\xrightarrow[(2)\text{湿Ag}_2\text{O}]{(1)\text{CH}_3\text{I}} \xrightarrow{\Delta} ? \xrightarrow[(2)\text{湿Ag}_2\text{O}]{(1)\text{CH}_3\text{I}} ? \xrightarrow{\Delta}$
18.  $\xrightarrow[\Delta]{\text{P}_2\text{O}_5}$
19.  $+\text{(CH}_3\text{CH}_2\text{CO)}_2\text{O} \xrightarrow[\Delta]{\text{CH}_3\text{CH}_2\text{COONa}}$
20. $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO} \xrightarrow[(\text{CH}_3\text{CH}_2)_2\text{O}]{\text{Mg}} \xrightarrow{\text{H}_3\text{O}^+}$
21. $\text{CH}_3\text{CH}_2\text{COOH} \xrightarrow[\text{P}]{\text{Br}_2} ? \xrightarrow{\text{NH}_3}$
22.  $\xrightarrow{\text{C}_2\text{H}_5\text{MgBr}} \xrightarrow{\text{H}_3\text{O}^+}$

三、用简单的化学方法鉴别下列各组化合物(8分)

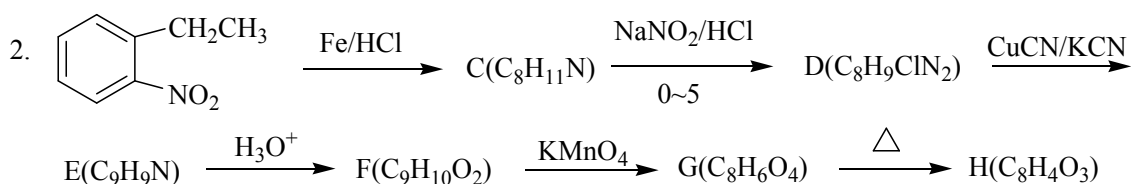
1. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ $\text{CH}_3\text{COCH}_2\text{CH}_3$ $\text{CH}_3\text{CHOHCH}_2\text{CH}_3$



2.  a  b  c  d  e

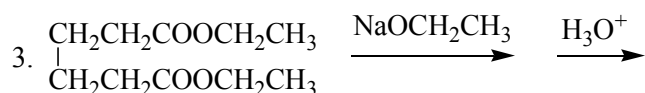
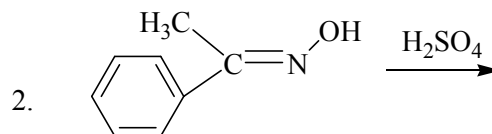
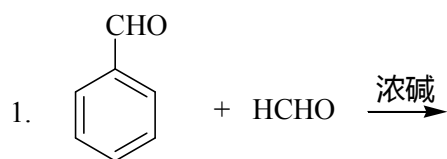
四、推测结构(10分)

1. 某化合物A分子式为 $\text{C}_7\text{H}_{12}\text{O}_3$,具有旋光性,A可与2,4-二硝基苯肼作用生成黄色结晶,并能使溴水褪色。A经稀氢氧化钠溶液水解后酸化,加热,放出二氧化碳气体,在所余溶液中加入 I_2/NaOH 可得碘仿沉淀和一化合物B($\text{C}_4\text{H}_7\text{O}_2\text{Na}$)。A与浓氢氧化钠溶液共热可生成一分子B及另一分子羧酸钠。写出A,B可能的结构式。

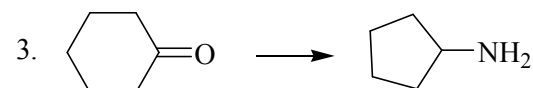
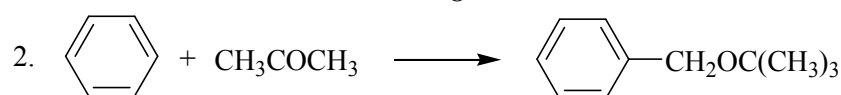
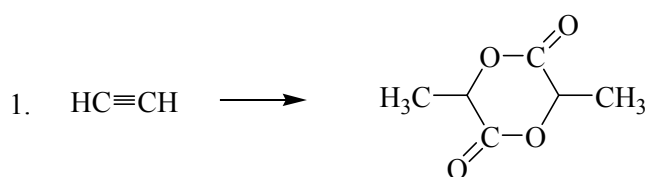


请写出C,D,E,F,G,H的结构式。

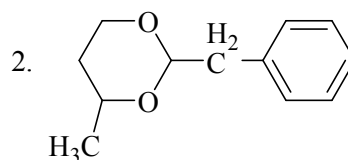
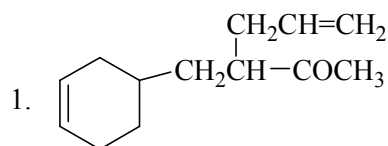
五、完成下列反应，并写出合理的反应历程(任选两题)(12分)



六、完成下列转变，其它试剂任选(任选两题)(12分)



七、由苯、甲苯及C3以下的有机原料合成下列化合物,无机试剂任选(任选三题)(18分)



(此题必做，且用乙酰乙酸乙酯的方法)

